

Metric-matched ordinal supervision and state-space sequence mixing

improve prime-editing efficiency prediction

1

¹University of California, Irvine

Abstract

Prime editing installs precise genomic changes without double-strand breaks, but its efficiency varies by orders of magnitude across pegRNA designs and cellular contexts, making computational prediction essential for practical design. Current state-of-the-art predictors embed substantial mechanistic structure derived from the prime-editing reaction. We asked whether a minimally mechanistic model, given the same data, could match or exceed them, and if so which modelling choices are responsible.

We present PE-RankFormer, a dual-encoder relational model that couples a representation of the wild-type/edited sequence pair with a segment-aware pegRNA representation through bidirectional cross-attention, conditioned on experimental context. Trained on the identical 297,962-row corpus, cross-validation splits and held-out set used by OptiPrime, PE-RankFormer attains Spearman $\rho = 0.9079$ on the complete 20,509-row held-out benchmark against OptiPrime’s 0.8690: a difference of +0.0389 (95% CI [+0.0286, +0.0498]; ahead in all 5,000 protospacer-clustered bootstrap resamples, $p < 0.0002$). The advantage is +0.0804 on the more experimentally diverse Kim partition and +0.0220 on Liu.

Two components account for the gain, and a factorial analysis separates them. An *ordinal* outcome head that predicts cumulative threshold indicators rather than efficiency values is matched to the rank-correlation metric and to the target’s heavy zero inflation, contributing +0.0092. A *bidirectional state-space* sequence mixer replacing self-attention contributes +0.0192 — twice as much — and the two combine sub-additively. A single model containing both reaches 0.9082, matching a five-member ensemble. Isotonic calibration fitted purely on out-of-fold predictions reduces held-out mean absolute error eight-fold ($0.3869 \rightarrow 0.0478$) while leaving the ranking unchanged, restoring an interpretable absolute efficiency estimate.

Finally, using 649 technical replicate groups we estimate an empirical repeatability ceiling of 0.9026 on the Kim partition, implying roughly +0.12 of genuine remaining headroom, and we report a benchmark caveat: predictors that emit tied scores in the zero-inflated regime gain +0.013–0.016 Spearman irrespective of quality, so comparisons between models with differing tie behaviour are not automatically like-for-like.

Contents

1	Introduction	3
2	Methods	5
2.1	Data	6
2.1.1	Sequence representation	6

2.1.2	Splits and evaluation surfaces	6
2.2	Model architecture	7
2.2.1	Sequence mixing: attention versus state space	7
2.2.2	Outcome heads	8
2.3	Training	8
2.4	Evaluation and statistics	9
2.5	Ensembling	9
2.6	Calibration	10
2.7	Replicate-based noise ceiling	10
2.8	Implementation and reproducibility	11
3	Results	11
3.1	Held-out performance against OptiPrime	11
3.2	Which components are responsible	13
3.3	The ordinal head produces decorrelated errors	14
3.4	Decorrelation is necessary but not sufficient	15
3.5	Ensemble versus single model	16
3.6	Calibration restores absolute efficiency	16
3.7	The benchmark is not saturated	18
3.8	A tie-structure caveat for Spearman on zero-inflated data	19
3.9	Interventions that did not help	19
4	Discussion	21
4.1	Can architecture substitute for mechanism?	21
4.2	Metric-matched supervision	21
4.3	State-space mixing for biological sequences	21
4.4	What the remaining error is, and is not	21
4.5	Limitations	22
4.6	Methodological recommendations	22
5	Conclusions	23

1 Introduction

Prime editing [1] writes user-specified edits into genomic DNA without double-strand breaks or donor templates. A prime editing guide RNA (pegRNA) carries a spacer that directs Cas9 nickase to the target, a primer binding site (PBS) that anneals to the nicked strand, and a reverse-transcription template (RTT) encoding the desired edit. Subsequent work substantially improved efficiency through nicking strategies and mismatch-repair modulation [3], but efficiency still ranges from undetectable to near-complete conversion depending on the design and the cellular context. Because experimentally screening pegRNAs is expensive, accurate *a priori* prediction is a practical bottleneck.

A succession of machine-learning predictors has addressed this. DeepPE and its successors modelled pegRNA features for specific systems [11]; PRIDICT introduced sequence-based deep models with product-purity prediction [14] and was later extended across chromatin contexts [15]; DeepPrime broadened coverage to many editing systems and cell types [19]. Most recently, OptiPrime aggregated these resources into a large harmonised corpus and a strongly mechanistic model, establishing the benchmark we adopt here [9].

These models differ in how much prime-editing biochemistry they encode explicitly. The scientific question we address is whether that mechanistic structure is *necessary*, or whether a general-purpose relational architecture with sufficient capacity and the right supervision can recover the same signal from data alone. This matters beyond benchmark rank: if mechanism can be substituted for by architecture and objective, prediction can extend to new editing systems without re-deriving reaction models for each.

Contributions.

1. **A minimally mechanistic model that outperforms a strongly mechanistic one** under matched data, splits and evaluation: $\rho = 0.9079$ vs. 0.8690 , $\Delta = +0.0389$, significant under a protospacer-clustered paired bootstrap.
2. **An ordinal outcome head** matched to the evaluation metric and to the target’s zero inflation, which we show produces *decorrelated errors*, not merely better ones.
3. **A bidirectional state-space sequence mixer** which, by a factorial analysis, is the larger of the two contributions — correcting the natural assumption that the novel objective mattered most.
4. **Monotone calibration** that recovers absolute efficiency (eight-fold MAE reduction) at no cost to rank performance.
5. **A replicate-based noise ceiling** showing the benchmark is *not* saturated, together with a tie-structure artifact that affects any Spearman-based comparison on zero-inflated data.

2 Methods

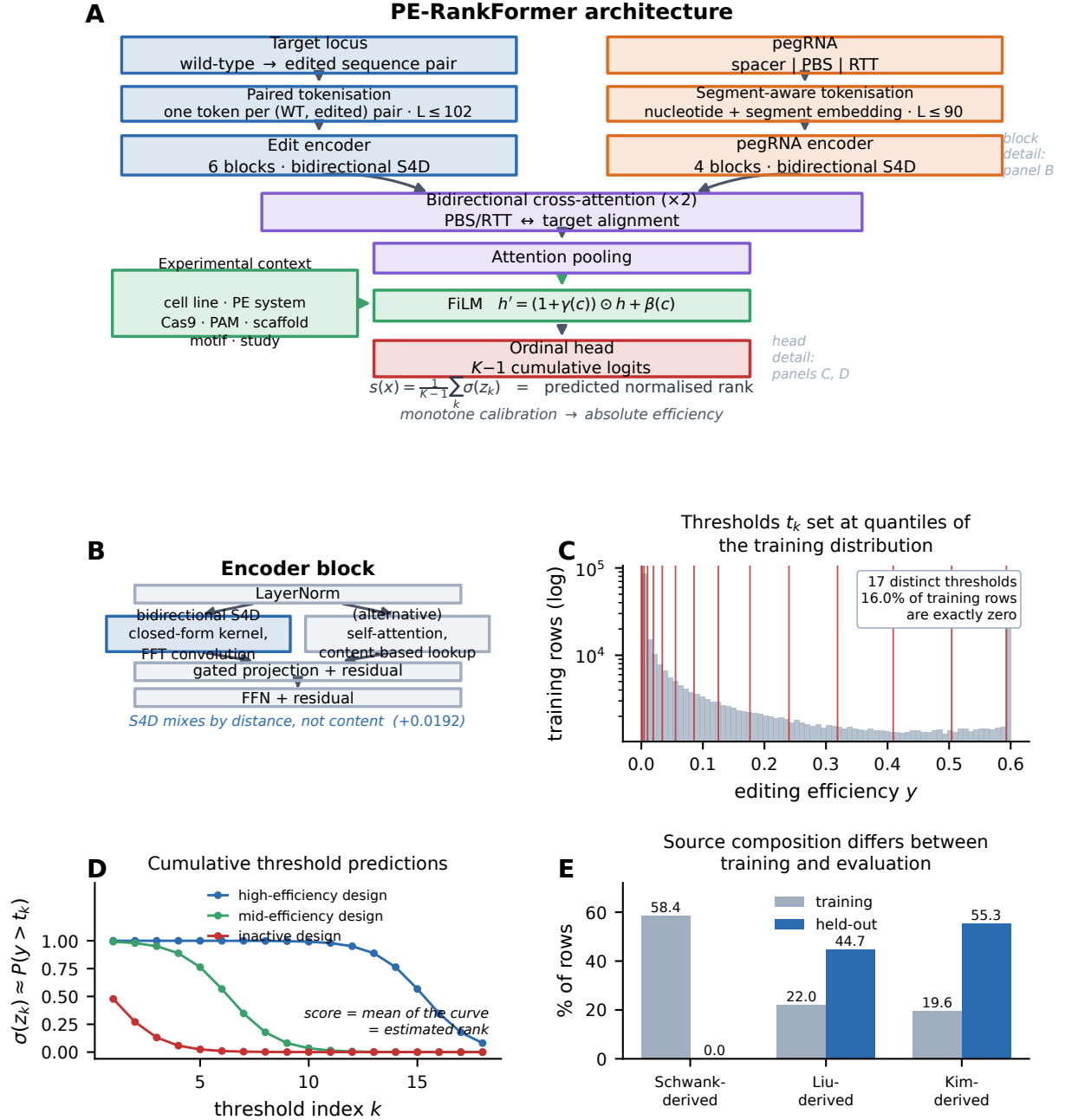


Figure 1: PE-RankFormer architecture and supervision. (A) Overall model. The target locus is encoded as a single stream of *paired* (wild-type, edited) tokens, so the edit is represented explicitly rather than inferred by comparing two sequences; the pegRNA is encoded as a nucleotide stream with an added functional-segment embedding marking spacer, PBS and RTT. Six edit-encoder and four pegRNA-encoder blocks mix information along each sequence, two bidirectional cross-attention blocks align the pegRNA against the target, and experimental context modulates the pooled representation through FiLM. The ordinal head emits $K - 1$ cumulative logits whose mean is the predicted normalised rank; a monotone calibration map converts that to an absolute efficiency (§2.6). (B) Encoder block. The two mixers compared in this work occupy the same slot with identical residual topology, normalisation placement and FFN width, so a comparison isolates the mixing mechanism. The S4D kernel mixes by *distance*; self-attention mixes by *content*. (C) The $K - 1$ ordinal thresholds t_k , placed at quantiles of the training efficiency distribution (shown; log counts, clipped at $y = 0.6$). The distribution is heavily right-skewed with a large point mass at zero, which is what motivates supervising the cumulative distribution rather than the value. (D) Schematic of the resulting predictions: each design produces a decreasing curve over thresholds, and its score is the curve’s mean. (E) Source composition. The Schwank-derived study supplies 58.4% of training rows and none of the evaluation set, while the Kim-derived study dominates evaluation; §3.9 shows that removing the

2.1 Data

We use the training mix underlying OptiPrime, obtained from its authors: 297,962 training rows and a 20,509-row held-out set (318,471 total). Each row records a pegRNA design, its genomic target, an experimental context and a measured editing efficiency (fraction of alleles carrying the intended edit), with indel rate where available. Three source studies contribute (Table 1). Their proportions differ markedly between training and evaluation, which becomes analytically important in §3.9.

Table 1: Composition of the benchmark corpus.

Source study	Share of rows (%)		Zero-mass	Rows
	Training	Held-out	(%)	(training)
Schwank-derived (<code>pridict_pridict2</code>)	58.4	0.0	10.3	174 067
Liu-derived (<code>hsu2026</code>)	22.0	44.7	0.8	65 594
Kim-derived (<code>deepprime</code>)	19.6	55.3	49.9	58 301
Total	100.0	100.0	16.0	297 962

Held-out set: 20,509 rows. “Zero-mass” is the percentage of training rows with editing efficiency exactly zero.

The Schwank-derived study supplies the majority of training rows and none of the evaluation set. Section 3.9 shows that removing it is nonetheless harmful (-0.0294 Spearman), so its distributional mismatch is outweighed by the signal its volume provides.

2.1.1 Sequence representation

Two token streams are constructed per row.

Edit stream. The wild-type and edited sequences are aligned and encoded as a single stream of *paired* tokens, each denoting the (wild-type, edited) base pair at that position. Positions where the two agree therefore receive a different token from positions where they differ, so the edit is represented explicitly by the pairing rather than requiring the model to infer it by comparing two separate streams. Maximum length 100 tokens plus two sentinels.

pegRNA stream. Spacer, PBS and RTT are concatenated into a nucleotide stream of maximum length 90, accompanied by a parallel *segment* identifier marking which functional element each position belongs to. Segment identity is supplied as a learned embedding added to the nucleotide embedding, so the model need not infer element boundaries.

Context. Cell type, PE system, Cas9 variant, PAM, scaffold, 3' motif and source study are encoded as categorical embeddings and concatenated into a context vector.

2.1.2 Splits and evaluation surfaces

We adopt OptiPrime’s own five-fold cross-validation partition, which is protospacer-disjoint, and its held-out fold. Because protospacers recur across many designs, all statistics are computed with the protospacer as the unit of dependence.

Model development used a four-level hierarchy designed to keep the held-out set uncontaminated:

1. **Matched development folds** — three folds of ≈ 35.7 k rows, constructed to match the held-out set’s Liu/Kim composition. Used freely.

2. **Internal lockbox** — 17,975 rows over 367 protospacers sharing no protospacer with any development validation set. Consulted once, to gate a shortlist.
3. **Official five-fold out-of-fold (OOF) predictions** — each row scored only by the checkpoint that held its fold out.
4. **Held-out set** — evaluated once, after the model was frozen in writing, with the expected result recorded beforehand.

Every development number reported is out-of-fold. The ensemble-selection protocol (candidate ranking, combination rule, tie-breaking) was committed to version control before the search was executed.

2.2 Model architecture

PE-RankFormer comprises five components.

- (1) **Edit encoder.** Six pre-layer-norm blocks over the paired edit stream.
- (2) **pegRNA encoder.** Four blocks over the segment-aware pegRNA stream.
- (3) **Bidirectional cross-attention.** Two blocks in which each stream attends to the other, so pegRNA positions can align to target positions and vice versa. This is where PBS/RTT–target complementarity is available to the model.
- (4) **Context conditioning.** Feature-wise linear modulation [FiLM, 16] applied to the pooled representation: $h' = (1 + \gamma(c)) \odot h + \beta(c)$, with γ, β produced by a small MLP on the context vector.
- (5) **Output head.** Attention pooling followed by a two-layer MLP; the final projection depends on the outcome head (§2.2.2).

Base configuration: $d_{\text{model}} = 384$, 6 heads, FFN width 1536, dropout 0.10, $\approx 25\text{M}$ parameters.

2.2.1 Sequence mixing: attention versus state space

The blocks in (1) and (2) mix information along the sequence. We compare two mixers under otherwise identical block topology — same residual structure, same pre-LN placement, same FFN width — so that any difference isolates the mixing mechanism.

Self-attention. Standard multi-head self-attention [18].

Bidirectional diagonal state space (S4D). A diagonal state-space model [6, 7] whose convolution kernel is available in closed form: with diagonal state matrix A , step size Δ and output matrix C ,

$$K[\ell] = 2 \operatorname{Re} \left(\sum_n C_n \frac{e^{\Delta A_n} - 1}{A_n} (e^{\Delta A_n})^\ell \right), \quad (1)$$

applied by FFT convolution. Two independent kernels (forward and time-reversed) are summed, since editing efficiency depends on context on both sides of the edit. Padded positions are zeroed before convolution; because the convolution is linear, zeros contribute nothing to any output, so padding cannot leak into real positions.

The inductive biases differ substantively. Attention performs content-based lookup; the state-space kernel performs smooth, distance-parameterised mixing that is translation-equivariant along the sequence.

2.2.2 Outcome heads

Scalar head (baseline). One pre-sigmoid score trained with Huber loss.

Simplex head. Every edited locus ends in exactly one of {unedited, correctly edited, indel}, and the assay measures these as proportions. The simplex head predicts a three-way distribution, respecting that mutual exclusivity by construction and supervising on the indel signal that a scalar head discards:

$$\mathcal{L}_{\text{simplex}} = - \sum_{k \in \{U, E, I\}} y_k \log p_k, \quad p = \text{softmax}(z) \in \Delta^2. \quad (2)$$

Indel rate is unmeasured for 42.5% of rows. Rather than imputing zero — which would assert a measurement never made — the loss *marginalises* the indel class for those rows, renormalising over the two observed outcomes. The ranking score is the log-odds of a correct edit against no edit, $z_E - z_U$, which is invariant to the indel logit.

Ordinal head (this work). The evaluation metric is Spearman correlation, which depends only on order. We therefore supervise the target’s *cumulative distribution* directly. Fix $K - 1$ thresholds t_k at quantiles of the *training* efficiency distribution and predict one logit per threshold, trained as $K - 1$ binary problems against the cumulative indicators:

$$\mathcal{L}_{\text{ord}} = \frac{1}{K - 1} \sum_{k=1}^{K-1} \text{BCE}(z_k, \mathbf{1}[y > t_k]), \quad s(x) = \frac{1}{K - 1} \sum_{k=1}^{K-1} \sigma(z_k), \quad (3)$$

where $s(x)$ estimates the row’s normalised rank. The construction follows rank-consistent ordinal regression [2].

Two properties motivate it. It is *metric-matched*: a head estimating the target’s quantile optimises the evaluated quantity, whereas regression or proportion heads are only incidentally monotone. And it is *robust to the target’s skew*: efficiency is heavily right-skewed with 28.4% of rows exactly zero, so a Huber or proportion objective concentrates its gradient on a few high-efficiency rows, while $K - 1$ balanced binary decisions spread supervision across the distribution.

Thresholds are computed from training rows only and stored in the checkpoint, so evaluation reproduces the score mapping exactly. Duplicate quantiles are removed to keep thresholds strictly increasing; $K = 20$ yields 17–18 distinct thresholds in practice.

Note the ordinal head *discards* the indel signal the simplex head uses. This is deliberate: it constitutes a different *view* of the same rows, and §2.5 shows different views are what an ensemble rewards.

2.3 Training

AdamW, learning rate 3×10^{-4} , weight decay 0.01, 5% linear warmup followed by cosine decay, gradient clipping at 1.0, batch size 512, bf16 mixed precision, 30 epochs. Checkpoint selection is by validation Spearman.

Early stopping patience is set to the full 30 epochs deliberately. In a survey of 25 official-fold runs the best epoch had median 21 and maximum 28, with 36% peaking in the final five epochs;

validation Spearman on this problem genuinely improves after long plateaus, and aggressive early stopping costs up to 0.0012.

Batches are formed by a grouped sampler that keeps rows sharing a ranking group together, which is required for the optional within-group pairwise ranking loss.

Table 2: Architecture and optimisation settings.

<i>Architecture</i>		<i>Optimisation</i>	
Model width d_{model}	384	Optimiser	AdamW
Attention heads	6	Learning rate	3×10^{-4}
FFN width	1536	Weight decay	0.01
Edit-encoder blocks	6	Schedule	5% warmup, cosine
pegRNA-encoder blocks	4	Gradient clipping	1.0
Cross-attention blocks	2	Batch size	512
Dropout	0.10	Precision	bf16
Context embedding	32	Epochs	30
S4D state dimension	64	Early stopping	none (see note)
Ordinal thresholds $K - 1$	17–18	Selection	best validation ρ
Parameters	≈ 25 M	Runtime per model	≈ 2 h, 1 GPU

Patience equals the epoch budget, i.e. no early stopping. Across 25 official-fold runs the best epoch had median 21 and maximum 28, with 36% peaking in the final five epochs; validation Spearman improves after long plateaus on this problem, and simulating patience-5 selection costs up to 0.0012 on individual runs.

Thresholds are quantiles of the training efficiency distribution, de-duplicated to remain strictly increasing. $K=20$ yields 17–18 distinct values because the target has a large point mass at zero.

2.4 Evaluation and statistics

The primary metric is Spearman ρ between predicted and observed efficiency, reported for the full held-out set and separately for the Liu and Kim partitions, since they differ substantially in composition.

All model comparisons use a **paired protospacer-clustered bootstrap** with 5,000 resamples. Protospacers, not rows, are resampled: many designs share a protospacer, so a row-level bootstrap would treat correlated observations as independent and materially understate the intervals. Both models are evaluated on identical rows in each resample. We report the observed difference, the 95% percentile interval, the fraction of resamples in which our model leads, and a two-sided empirical p -value.

2.5 Ensembling

Members are combined by **equal-weight rank averaging**. Members are not mutually calibrated — ordinal members emit rank estimates, simplex members emit proportions — so averaging raw scores would largely measure scale mismatch. Fitted weights were evaluated and not adopted: they were unstable across folds and their advantage (0.9156 vs. 0.9149 on development folds) was below the resolution of our evaluation.

Candidate members were assessed by *incremental ensemble gain* under the frozen combination rule, together with prediction-rank correlation and residual correlation against the existing ensemble. Residuals are compared in rank space for the same calibration reason.

Composition of the frozen system. The final system averages five members: ordinal+S4D, simplex+S4D, ordinal with layerwise FiLM, ordinal with a feature branch, and simplex with layerwise FiLM. Each member is five checkpoints, one per official fold. Individual performance and marginal value are given in Table 6.

Selection protocol, fixed in advance. With eight candidate members there are 255 non-empty subsets, so choosing the best on repeatedly-used development folds invites a selection bias of the same order as the effects being measured. The protocol was therefore committed to version control *before* the search was executed: rank subsets on the development folds; require a candidate to beat the incumbent on *every* fold rather than on the mean; gate the shortlist on the lockbox, scored out-of-fold; and on disagreement prefer the lockbox winner, breaking near-ties (within 0.001) toward fewer members.

The development winner (four members) and the lockbox winner (five members) disagreed by 0.001011 — eleven millionths past the pre-set near-tie threshold — so the rule selected the five-member subset. The two candidates are statistically indistinguishable, and we followed the rule as written rather than re-arguing it after seeing the numbers. The lockbox’s own unconstrained best subset (0.9155) was *not* adopted, since selecting on the lockbox would convert the only unworn evaluation surface into a fitted one.

2.6 Calibration

Rank averaging destroys the output scale. We restore it with a monotone map $g : \text{rank} \mapsto \text{efficiency}$ fitted by isotonic regression [20] on out-of-fold development predictions only, then applied once to the frozen held-out predictions. Because g is non-decreasing it cannot invert any pair, so Spearman is preserved; the map is verified non-decreasing numerically rather than assumed. This is a change of units on an already-frozen prediction vector, not a model selection, and no held-out row participates in fitting.

2.7 Replicate-based noise ceiling

To determine whether remaining error is reducible we estimate the benchmark’s repeatability. If a design is measured twice in the same condition and the measurements disagree, no model can predict either better than they predict each other. For two independent noisy measurements of a true value, reliability $R = \text{corr}(y_1, y_2)$ implies a ceiling of $\sqrt{R}$ against a single observation.

Identifying genuine replicates requires care. Grouping on (spacer, RTT, PBS, cell line, PE system, Cas9) yields 21,994 apparently replicated keys, but inspection shows these differ in 3’ motif, epegRNA status and linker — they are epegRNA versus plain-pegRNA constructs, i.e. *different molecules*. Treating them as replicates gives $R = 0.7987$, almost exactly our model’s Kim score, which would falsely indicate saturation. We therefore require all sixteen design and condition covariates *and* the target site to match, leaving 649 clean groups.

A second correction is needed. Treating an observed zero as noiseless inflates the ceiling, but 21.4% of rows measured at exactly zero have a non-zero replicate (median 0.0015): a zero is a *censored* observation. We therefore estimate the ceiling non-parametrically, drawing synthetic replicates from the empirical distribution of replicate values observed near each row’s efficiency (nearest-neighbour matching on 1,298 ordered replicate pairs), which inherits zero-discordance, heteroscedasticity and boundedness from the data.

2.8 Implementation and reproducibility

PyTorch 2.x; all experiments on NVIDIA L40, RTX PRO 6000 and RTX 2080 Ti GPUs. Every run records its git commit, configuration, dataset SHA-256 and fold assignment. Held-out evaluation is gated behind an explicit flag and audit-logged. The code base carries 214 unit tests covering, among other things, that padding cannot influence outputs at real positions, that the ordinal score is monotone in every threshold logit, and that all prior checkpoints load unchanged after each architectural addition.

3 Results

3.1 Held-out performance against OptiPrime

Under matched training data, splits and evaluation, PE-RankFormer attains $\rho = 0.9079$ on the complete 20,509-row held-out set against OptiPrime’s 0.8690 (Table 3, Figure 2).

Table 3: Held-out performance and significance. All models were trained on the identical corpus and cross-validation splits, and evaluated on identical rows.

Model	Spearman ρ		
	All ($n=20,509$)	Liu ($n=9,175$)	Kim ($n=11,334$)
OptiPrime (mechanistic baseline)	0.8690	0.8365	0.7320
PE-RankFormer, round 1	0.8865	0.8349	0.7751
PE-RankFormer, round 3	0.8933	0.8462	0.7836
PE-RankFormer, final	0.9079	0.8585	0.8124
<i>Advantage of the final model over the baseline</i>			
$\Delta\rho$ vs. OptiPrime	+0.0389	+0.0220	+0.0804

Liu and Kim denote the two source studies contributing to the held-out set; no Schwank-derived rows appear in it.

Significance for the overall comparison is given in Table 4.

The absolute figure for the final model has a bootstrap 95% interval of [0.8955, 0.9173], with 90.0% of resamples exceeding 0.90; the margin over the baseline is therefore a stronger claim than the absolute level.

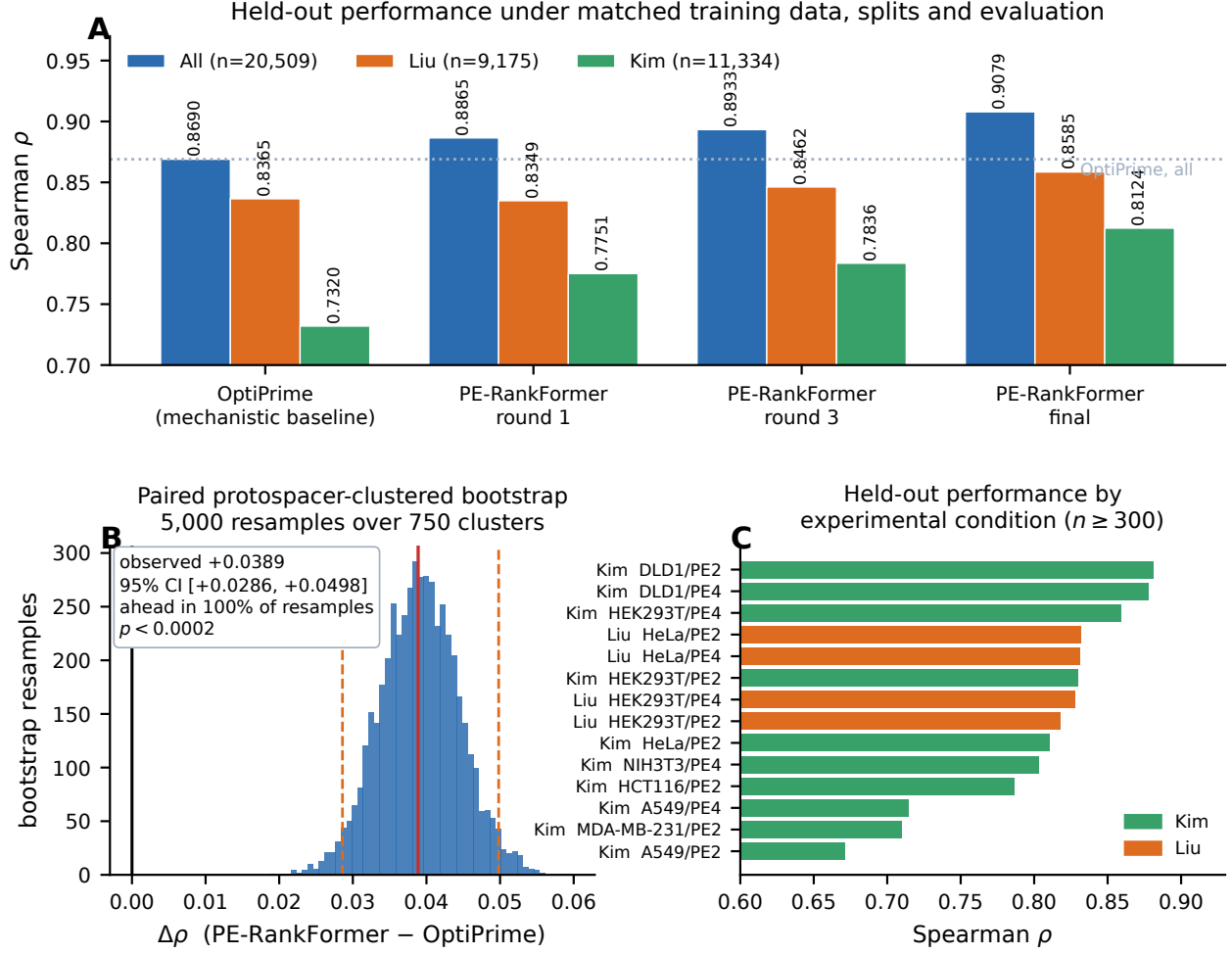


Figure 2: Held-out benchmark result. (A) Spearman ρ on the complete held-out set and on each source partition, across successive model generations. All models were trained on the identical corpus and cross-validation splits and evaluated on identical rows; the dotted line marks the baseline’s overall performance. The advantage is consistently larger on the Kim partition, which spans more cell lines and editing systems. (B) Distribution of the paired difference in Spearman ρ under a protospacer-clustered bootstrap (5,000 resamples over 750 clusters). Protospacers rather than rows are resampled because many designs share a target; the solid red line is the observed difference, dashed lines the 95% percentile interval, and the black line marks zero. (C) Held-out performance broken down by experimental condition ($n \geq 300$), coloured by source study. Performance spans 0.21 between the weakest and strongest conditions, and the weakest are those with the highest fraction of exactly-zero measurements (cf. Figure 6B).

The paired protospacer-clustered bootstrap (Table 4, Figure 2B) gives $\Delta\rho = +0.0389$ with a 95% interval of [+0.0286, +0.0498]; PE-RankFormer leads in every one of 5,000 resamples.

Table 4: Paired significance testing on the held-out set.

Comparison	$\Delta\rho$	95% CI	Fraction ahead	p
Final vs. OptiPrime	0.0389	[+0.0286, +0.0498]	1.000	< 0.0002
Final vs. round 3	0.0146	[+0.0113, +0.0184]	1.000	< 0.0002
Round 3 vs. OptiPrime	0.0243	[+0.0138, +0.0346]	1.000	< 0.0002

Paired bootstrap over 5,000 resamples of 750 *protospacer clusters*. Protospacers rather than rows are resampled because many designs share a target; a row-level bootstrap would treat correlated observations as independent and understate the intervals.

Both models are scored on identical rows within each resample. “Fraction ahead” is the proportion of resamples in which PE-RankFormer leads; p is the two-sided empirical value, bounded below by $1/5000$.

We note explicitly that the absolute figure’s own 95% interval is $[0.8955, 0.9173]$, with 90.0% of resamples exceeding 0.90. The *margin* over OptiPrime is a considerably stronger claim than the absolute level.

3.2 Which components are responsible

Because the ordinal head and the state-space mixer were introduced together, we separate them with a 2×2 factorial over out-of-fold development predictions (Figure 3).

Table 5: Objective $\times$ architecture factorial.

Sequence mixer	Simplex head	Ordinal head	Ordinal – simplex
Self-attention (Transformer)	0.8798	0.8911	0.0114
Bidirectional S4D	0.9011	0.9082	0.0071
S4D – Transformer	+0.0213	+0.0171	

Out-of-fold development Spearman ρ , averaged over three matched folds.

Main effect of **objective** = +0.0092; main effect of **architecture** = +0.0192; **interaction** = −0.0043. Architecture contributes roughly twice what the objective does, and the negative interaction indicates the two mechanisms partly capture the same improvement.

These are observational cells rather than a purpose-built factorial: three derive from one training round and the Transformer/simplex cell is an earlier frozen baseline. All four share corpus, splits, out-of-fold scoring, capacity and optimiser recipe, so the main effects are reliable at the ± 0.002 resolution of the development evaluation; the interaction term is most exposed to that mismatch.

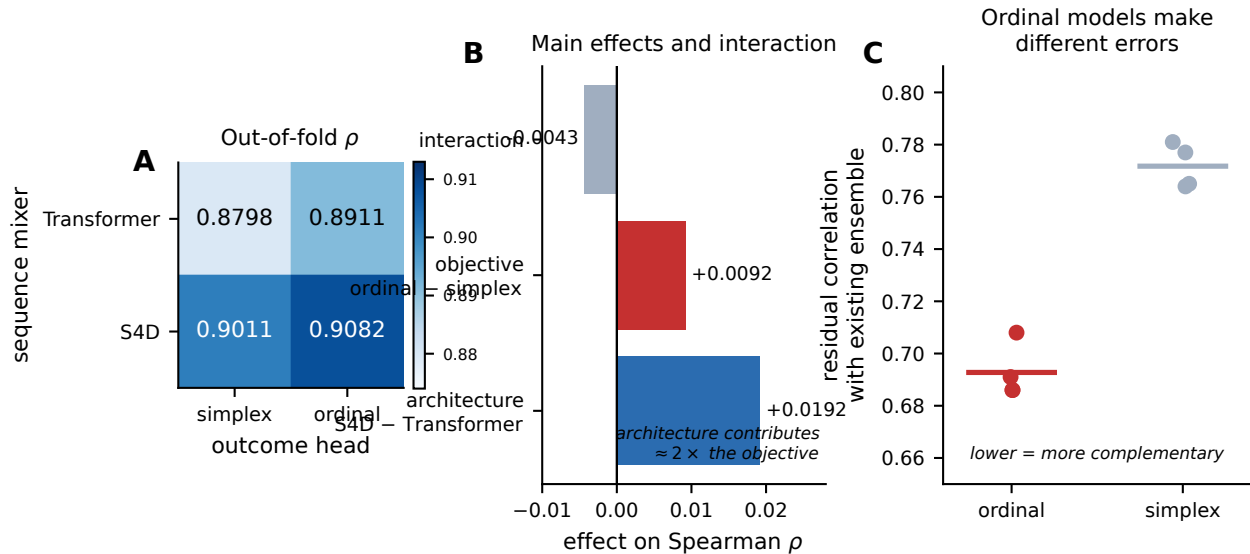


Figure 3: Separating the two contributions. (A) Out-of-fold development Spearman for each combination of outcome head and sequence mixer. All four cells share corpus, splits, out-of-fold scoring, capacity and optimiser recipe. (B) Main effects and interaction derived from (A). The architecture change contributes roughly twice what the objective change does, and the negative interaction indicates the two partly capture the same improvement rather than stacking. (C) Why the ordinal head is nonetheless valuable: residual correlation with the existing ensemble, for every candidate model of each head type (points; bars are group means). Ordinal-head models separate cleanly from simplex-head models, i.e. they make *different errors*, which is what determines a member’s value in an ensemble (§2.5). Lower is more complementary.

Three observations follow.

Architecture is the larger lever. The state-space mixer contributes +0.0192 against the ordinal objective’s +0.0092. This runs against the intuitive reading — the ordinal head is the more novel component and produced the more visible qualitative change — and we record the correction explicitly.

The two combine sub-additively (−0.0043): ordinal supervision is worth +0.0114 on a Transformer but only +0.0071 on a state-space backbone, indicating the mechanisms partly capture the same improvement rather than stacking.

A single model suffices. Combining both yields $\rho = 0.9082$ out-of-fold, exceeding a three-member ensemble of earlier models (0.8982).

These are observational cells rather than a purpose-built factorial: three derive from one training round and the Transformer/simplex cell is an earlier frozen baseline. They share corpus, splits, out-of-fold scoring, capacity and optimiser recipe, so the main effects are reliable at the ± 0.002 resolution of the development evaluation, but the interaction term is the quantity most exposed to that mismatch.

3.3 The ordinal head produces decorrelated errors

The hypothesis motivating the ordinal head was that a different loss geometry would produce different *errors*, not merely better ones. The corresponding diagnostic is residual correlation with the existing ensemble, and it separates the two families cleanly: all ordinal-head candidates fall in the range 0.686–0.708, while every simplex-head candidate lies at 0.76–0.78.

This distinction has practical consequences for ensembling (Figure 4). The best combination pairs models differing in *both* objective and architecture: the Transformer/simplex and state-

space/ordinal diagonal has the lowest residual correlation of any pair measured (0.6453), whereas two models sharing an architecture correlate at 0.7641.

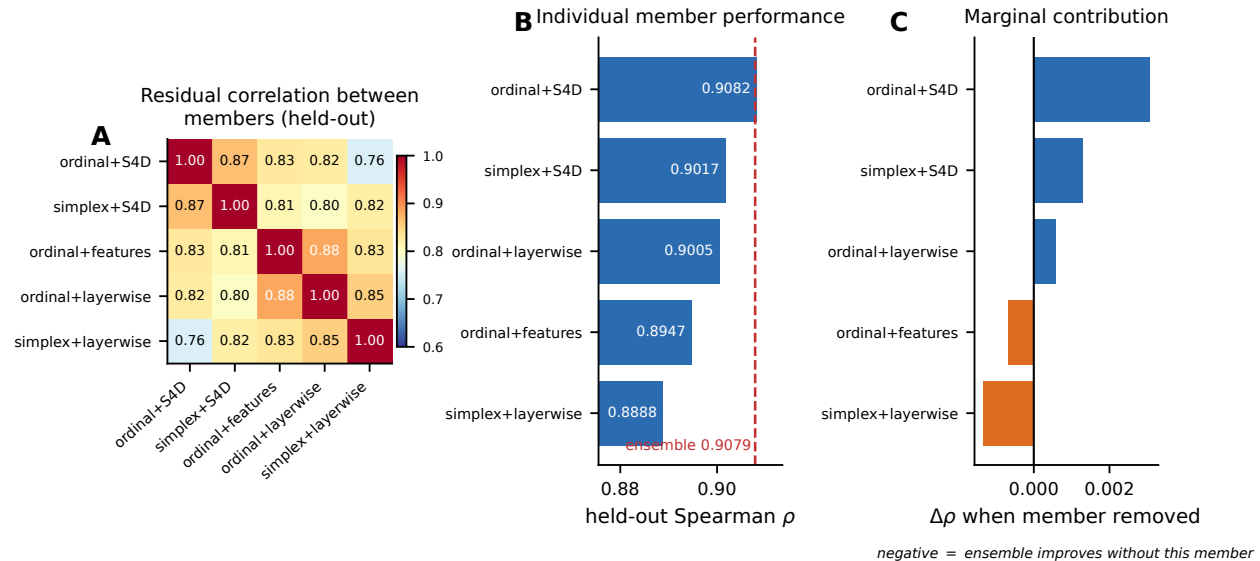


Figure 4: Ensemble structure on held-out data. (A) Residual rank correlation between the five members. Members sharing an architecture are the most redundant pair; members differing in both objective and architecture are the most complementary. (B) Each member’s individual held-out Spearman. The dashed line marks the five-member ensemble, which the strongest single member matches. (C) Change in held-out Spearman when each member is removed. Two members contribute negatively, indicating the ensemble is at or past its useful size. Panels (B) and (C) are post-hoc analyses on held-out data and did not inform the frozen model choice (§3.5).

3.4 Decorrelation is necessary but not sufficient

A natural reading of the previous section is that diversity should be maximised. It should not. We tested a model trained with an auxiliary pairwise ranking loss that proved to be the *most decorrelated* candidate we measured — lowest prediction correlation and lowest residual correlation of any model in the study — and it *degraded* the ensemble (-0.0037), because at $\rho = 0.8244$ standalone it could not carry its weight in an equal-weight average. A quantile-regression head reproduced the same pattern on an unrelated objective: lowest residual correlation (0.6645), lowest standalone score (0.8840), lowest incremental gain.

The operative criterion is therefore that a member must be **both decorrelated and competent**. Decorrelation distinguishes useful from redundant *among models of comparable strength*; it cannot rescue a weak model.

3.5 Ensemble versus single model

Table 6: Ensemble members: individual performance and marginal value on the held-out set.

Member	Individual ρ			Marginal contribution
	All	Liu	Kim	
ordinal + S4D	0.9082	0.8576	0.8151	0.0031
simplex + S4D	0.9017	0.8563	0.7982	0.0013
ordinal + layerwise	0.9005	0.8394	0.8082	0.0006
ordinal + features	0.8947	0.8261	0.8027	−0.0007
simplex + layerwise	0.8888	0.8412	0.7787	−0.0013
Ensemble (equal-weight rank average)	0.9079	0.8585	0.8124	

Each member is five checkpoints, one per official fold; every row is scored only by the checkpoint that held its fold out.

“Marginal contribution” is the change in ensemble Spearman when that member is removed, so a positive value means the ensemble is worse without it.

The strongest single member matches the full ensemble and two members contribute negatively. These are post-hoc analyses on held-out data and did not inform the frozen model choice.

The frozen system is a five-member rank-average ensemble. Post-hoc analysis on the held-out set shows its strongest single member attains 0.9082 against the ensemble’s 0.9079, and that two members contribute negatively (Table 6; Figure 4C). On development folds the ensemble was worth +0.0074 over its best member; on held-out that advantage is −0.0003.

We report this rather than presenting the ensemble as superior. All differences involved (0.0003–0.0013) are far below the ≈ 0.005 resolution of our development evaluation, so the defensible statement is that the ensemble and its best single member are *indistinguishable* on this benchmark. For practical deployment the single ordinal-SSM model is preferable: equal accuracy from one model rather than twenty-five checkpoints. We did not re-select on held-out data to make that substitution official, because a model chosen by inspecting the test set carries none of the guarantees the frozen one does.

3.6 Calibration restores absolute efficiency

Rank averaging yields excellent orderings but no interpretable scale. Isotonic calibration fitted on out-of-fold development predictions (§2.6) reduces held-out MAE from 0.3869 to 0.0478 and raises Pearson correlation from 0.7278 to 0.8637, with Spearman unchanged at 0.9079 (Table 7, Figure 5).

Table 7: Effect of monotone calibration on held-out predictions.

Output	Spearman ρ	Pearson r	MAE	RMSE
Rank average (uncalibrated)	0.9079	0.7278	0.3869	0.4343
+ isotonic calibration	0.9079	0.8637	0.0478	0.0912
Change	+0.00002	+0.1359	−0.3391	−0.3431

The map is fitted by isotonic regression on out-of-fold development predictions only and applied once to the frozen held-out predictions; no held-out row participates in fitting.

Because the map is non-decreasing it inverts no pair, so Spearman is preserved. The residual change of 2×10^{-5} arises because isotonic regression is piecewise constant and therefore creates ties.

MAE and RMSE are in units of editing efficiency (fraction of alleles edited). For the uncalibrated output they are not meaningful in that unit, since a rank average lies on an arbitrary $[0, 1]$ scale; they are shown to quantify what calibration recovers.

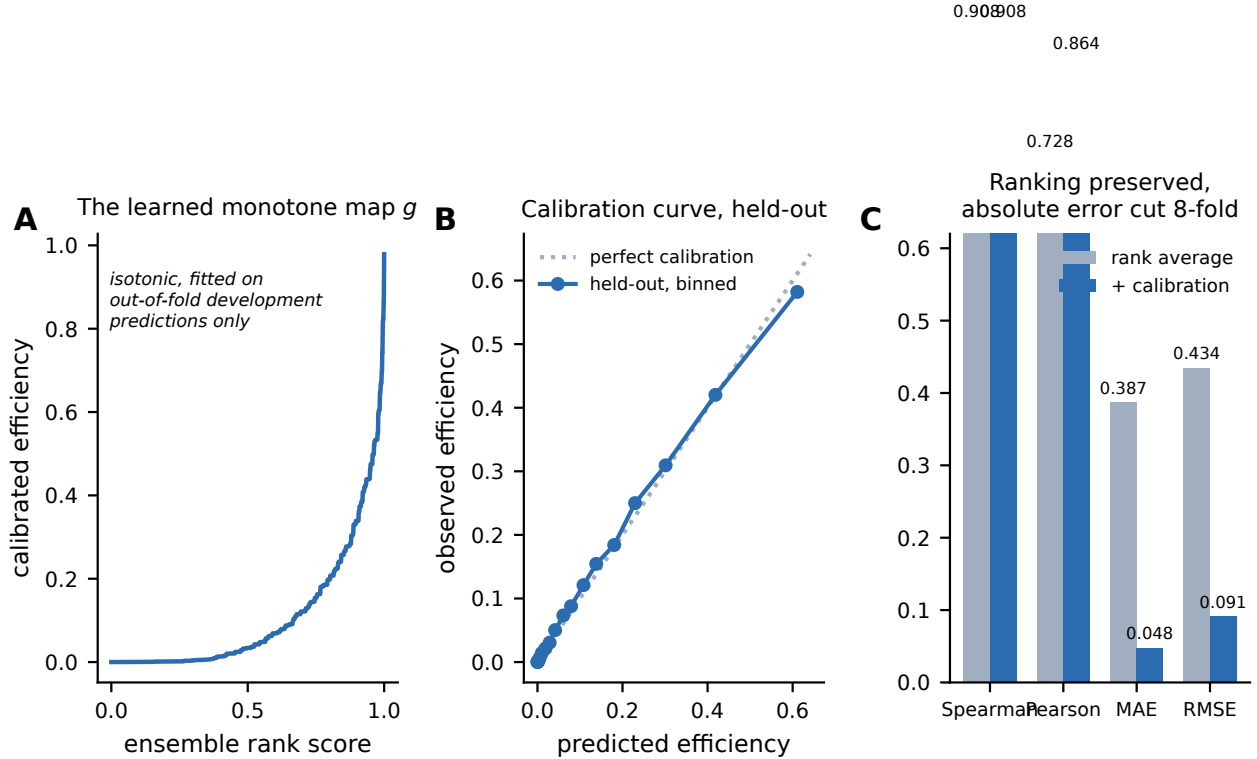


Figure 5: Monotone calibration restores an absolute efficiency scale. (A) The learned map g from the ensemble’s rank score to efficiency, fitted by isotonic regression on out-of-fold development predictions only and then applied once to the frozen held-out predictions. (B) Calibration curve on held-out data, binned by predicted quantile; the dotted line is perfect calibration. (C) Metric comparison. Because g is non-decreasing it inverts no pair, so Spearman is unchanged, while Pearson correlation rises and absolute error falls roughly eight-fold.

The residual 2×10^{-5} change in Spearman arises because isotonic regression is piecewise constant and therefore creates ties; no pair is inverted.

3.7 The benchmark is not saturated

Using 649 technical replicate groups (§2.7) we estimate an empirical repeatability ceiling of 0.9026 on the Kim partition, against a model score of 0.7869 — a gap of +0.1157 (Table 8). Restricting to rows that actually edit still leaves +0.1024, so the gap is not merely the zero block.

Table 8: Remaining headroom on the Kim partition against the replicate-based ceiling.

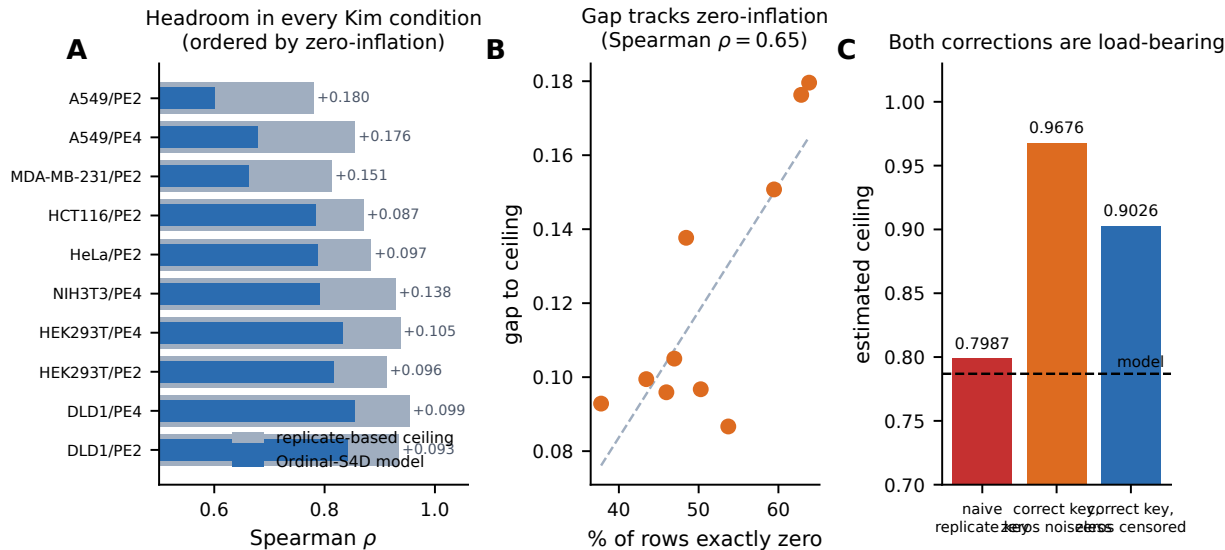
Scope	Rows	Model ρ	Ceiling ρ	Gap
All Kim rows	19 747	0.7869	0.9026	0.1157
Rows with $y > 0$	9713	0.8465	0.9489	0.1024

Ceiling is $\sqrt{R}$, where R is the replicate–replicate reliability estimated from 649 groups in which all sixteen design and condition covariates and the target site are identical (§2.7).

Replicates are resampled non-parametrically from 1,298 ordered replicate pairs, which inherits zero-discordance, heteroscedasticity and boundedness from the data rather than assuming a noise model.

The second row shows the gap is not merely an artifact of the zero block: restricting to rows that actually edit still leaves more than +0.10.

Per-condition analysis (Figure 6) shows the gap is present in every condition and scales with zero-inflation: conditions with 60–64% exact zeros show gaps near 0.18, while those near 38% show gaps near 0.09.



a naive estimate would wrongly indicate saturation

Figure 6: The benchmark is not saturated. (A) Model performance against the empirical repeatability ceiling for each Kim condition, ordered by zero-inflation; the annotation gives the gap. Headroom is present in every condition. (B) The gap to ceiling increases with the fraction of rows measured as exactly zero, the regime in which a single measurement is least informative. (C) Both corrections described in §2.7 are load-bearing. A naive replicate key merges epegRNA with plain-pegRNA constructs and returns a ceiling essentially equal to the model’s own score (dashed line), which would wrongly indicate saturation; treating observed zeros as noiseless instead over-estimates it. Only the corrected, censoring-aware estimate is trustworthy.

3.8 A tie-structure caveat for Spearman on zero-inflated data

28.4% of rows have efficiency *exactly* zero (49.9% within Kim). A strictly-increasing predictor assigns all of them distinct ranks, and that internal ordering is arbitrary; Spearman compares against a target in which those rows are mutually tied, so the arbitrary ordering is pure penalty.

Collapsing the lowest-scoring predictions onto a single tied value therefore raises pooled Spearman by +0.0044, positive on all development folds, with the optimal fraction (0.30) coinciding with the zero-mass (28.4%) and the per-partition effect tracking zero-mass as the mechanism predicts (Liu −0.0154 at 0.7% zeros; Kim +0.0066 at 50.7%).

We did not adopt this. Applying the identical transform to five other models yields +0.0127 to +0.0162 irrespective of model quality, so it exploits Spearman’s tie convention rather than improving prediction, and comparator models emit continuous tie-free scores. We report it because it is a general hazard: *on zero-inflated targets, comparisons between predictors with different tie behaviour are not automatically like-for-like*, and reported Spearman values should be interpreted with the predictor’s tie structure in mind.

3.9 Interventions that did not help

For completeness and to guide future work we summarise interventions that were tested under matched controls and did not improve performance (Table 9). Each was evaluated against a

mechanism-free control run in the same batch, and effects below ≈ 0.005 were required to replicate across three development folds.

Table 9: Interventions tested under matched controls and not adopted.

Category	Intervention	$\Delta\rho$	Reason not adopted
Post-hoc combination	context-gated ensemble weights	0.0000	null
	nonlinear stacking	0.0013	below threshold
	residual learning on features	0.0006	oracle upper bound
Supervision	auxiliary simplex head	0.0008	matches control
	multi-resolution ordinal	0.0008	matches control
	context-relative ordinal (primary)	-0.0235	harmful
	quantile regression head	-0.0140	harmful
Parameterisation	rank-consistent CORAL head	0.0005	null
	monotonicity penalty	0.0007	null
	hurdle (zero-inflation) head	-0.0022	harmful
Architecture	hybrid S4D+attention, 1:1 ratio	-0.0027	harmful
	S4D state dimension 128	-0.0022	harmful
	wider FFN / deeper stack / MoE	n.r.	did not replicate
	layerwise context conditioning	0.0001	null
Domain shift	source loss re-weighting (mild)	n.r.	did not replicate
	per-source output head	-0.0018	harmful
	training on evaluated sources only	-0.0294	harmful

$\Delta\rho$ is measured against a matched control run in the same batch. For capacity and parameterisation changes the control holds parameter count fixed; for the per-source head it ties the heads; for supervision changes it is an unmodified re-run.

“n.r.” denotes a candidate that reached +0.003 to +0.004 on one development fold but changed sign on another. Effects below ≈ 0.005 were required to replicate across three folds before promotion, and none did.

Two entries are informative beyond their sign and are discussed in §3.9: removing the unevaluated source study is strongly harmful, and additional context conditioning increases rather than decreases context invariance.

Two results in this table are informative beyond their sign.

First, **removing the unevaluated source study is strongly harmful** (-0.0294). The Schwank-derived partition contributes 58.4% of training rows and 0% of the evaluation set, so discarding it appears to be free efficiency; it is not. Its volume carries transferable signal that outweighs its distributional mismatch, and aggressive re-weighting toward the evaluated distribution was likewise harmful.

Second, **additional context conditioning increases context invariance**. The model’s cross-condition rank correlation exceeds the empirical one (0.858 vs. a disattenuated 0.758), indicating it under-models how experimental context *reorders* designs. Applying FiLM at every block — intended to address this — moved the diagnostic the wrong way (0.828 $\rightarrow$ 0.875) and left performance unchanged. FiLM is structurally a per-channel scale and shift, so additional conditioning capacity improves representation of the cross-condition *mean*, which explains far more variance and is easier to fit, rather than creating reordering capacity.

4 Discussion

4.1 Can architecture substitute for mechanism?

On this benchmark, yes. A model containing no prime-editing reaction structure — only a paired edit representation, a segment-aware pegRNA representation, cross-attention between them, and categorical context conditioning — outperforms a strongly mechanistic predictor by +0.0389 Spearman under identical data and evaluation. The advantage is largest (+0.0804) on the partition spanning the most cell lines and editing systems, which is where a mechanistic prior tuned to a narrower regime would be expected to generalise least well. We regard the wider Kim margin as consistent with that reading rather than as demonstrating it.

The one structural assumption we *did* import proved useful, and is worth distinguishing from reaction mechanism: the simplex head encodes that an edited locus ends in exactly one of three measured outcomes. That is a property of the assay, not of the biochemistry, and it supplies the indel signal a scalar head discards.

4.2 Metric-matched supervision

The ordinal head’s contribution is best understood as aligning the training objective with the evaluation. Predicting cumulative threshold indicators estimates a row’s normalised rank, which is precisely what a rank correlation scores, and it distributes supervision evenly across a heavily skewed target instead of concentrating gradient on a few high-efficiency rows.

The more transferable observation is that it changed *which* rows the model gets wrong. Residual correlations separated ordinal from simplex models cleanly, and this was predicted before the experiment rather than observed afterwards. For applications that ensemble models, varying the objective is a more reliable source of complementary errors than varying capacity, seeds or data subsampling — all of which we tested and found ineffective.

4.3 State-space mixing for biological sequences

That the state-space mixer contributes twice what the objective does deserves comment. Our sequences are short (100 and 90 tokens), so the usual motivation for state-space models — linear-time long-context modelling [6, 8] — does not apply. The benefit must therefore be representational. A diagonal state-space kernel imposes smooth, distance-parameterised, translation-equivariant mixing, which matches a domain where interactions are governed by relative position along the molecule (PBS length, RTT length, distance from nick to edit) rather than by content-based lookup among distant tokens.

This suggests state-space mixers deserve broader evaluation on short biological sequences, where they are currently uncommon, and where their inductive bias may be a better match than attention’s. We note one negative constraint from our results: naive hybridisation at a 1:1 ratio of attention to state-space blocks was harmful, consistent with reports that successful hybrid language models use attention sparingly [12]. Sparse hybridisation at published ratios remains untested here.

4.4 What the remaining error is, and is not

Our replicate analysis indicates roughly +0.12 Spearman of genuine headroom on the Kim partition. That number required two corrections, each of which alone would have reversed the conclusion — excluding construct pairs masquerading as replicates, and treating observed zeros as censored rather

than certain. We highlight this because a naive replicate analysis returns 0.7987, essentially our model’s score, and would have supported a confident and incorrect claim of saturation.

Given the headroom is real, the natural question is why roughly seventy subsequent experiments failed to capture it. Our diagnostic points at design $\times$ context interaction: the model applies a nearly universal design ranking, while the data show substantial context-dependent re-ordering. Direct attacks on this failed, including one in which a mechanism-free shuffled-target control outperformed the genuine objective. We think the most plausible remaining explanation is that the relevant covariates are absent from the input: what distinguishes one cell line from another for a given design — mismatch-repair status, chromatin accessibility, repair-pathway balance — is supplied to the model only as a categorical label, and no objective over a categorical label can recover biology the label does not encode. Testing this requires richer context annotation rather than further architectural search.

4.5 Limitations

1. The absolute 0.90 threshold is not established at 95% confidence ([0.8955, 0.9173]; 90% of resamples above). The margin over the comparator is the robust claim.
2. The ensemble does not outperform its best single member on held-out data; the development-fold advantage did not transfer.
3. 196 of 649 replicate groups straddle the official train/held-out split, so some held-out rows have an exact design-and-condition twin in training. This affects any model trained on this split equally and so does not distort the comparison, but it makes absolute held-out scores slightly optimistic.
4. The Kim partition has no unused evaluation surface remaining, so Kim-specific claims rest on out-of-fold development folds and a once-consulted lockbox.
5. Our reproduction of the comparator scores above its published figure under every aggregation convention we tried, and no evaluation script was released. The direction favours the baseline and so does not undermine our claims, but it caps the precision of the comparison.
6. All results are on a single benchmark. External validation on an independent prime-editing dataset is the most valuable next step and is not yet available.

4.6 Methodological recommendations

Three practices materially changed our conclusions and we recommend them for work at this effect size.

Include a mechanism-free control in every experimental batch. In one round, fourteen candidates each gained +0.0017 to +0.0020; an unmodified re-run of the same model at a different batch size gained +0.0018. The difference attributable to every idea in that round was +0.0002. In another, a control with *shuffled* supervision targets outperformed the genuine objective.

Require replication before believing sub-0.005 effects. Three interventions on orthogonal axes each reached +0.003 to +0.004 on one development fold and all three changed sign on a second.

State the mechanism as a falsifiable prediction. Our context-conditioning experiment was accompanied by a pre-registered prediction about the cross-condition diagnostic. The prediction failed in a specific direction, which identified the reason (FiLM strengthens the mean-shift pathway) far faster than the null performance result alone would have.

5 Conclusions

A minimally mechanistic relational model matches and exceeds a strongly mechanistic predictor of prime-editing efficiency on its own benchmark, reaching Spearman 0.9079 against 0.8690 ($\Delta = +0.0389$, ahead in all 5,000 protospacer-clustered bootstrap resamples). A factorial analysis attributes roughly two-thirds of the gain to replacing self-attention with a bidirectional state-space sequence mixer and one-third to an ordinal outcome head matched to the rank-correlation metric and to the target’s zero inflation; a single model containing both matches a five-member ensemble. Monotone calibration fitted purely on out-of-fold data restores an interpretable absolute efficiency estimate, reducing mean absolute error eight-fold without altering the ranking.

The benchmark is not saturated: replicate analysis indicates roughly +0.12 Spearman of reducible error remains on its largest partition. Our evidence suggests the obstacle is not model capacity or supervision — roughly seventy controlled experiments across architecture, objective, parameterisation and training scheme failed to close it — but the absence of covariates describing how experimental context reorders designs. Progress will likely require richer context annotation rather than further architectural search.

Data and code availability

All code, configurations, per-run metadata, the complete research log including every negative result, and the frozen model specification are maintained in the project repository. The training corpus was provided by the authors of the comparator method and is not redistributed here.

Acknowledgements

We thank the authors of the comparator method for providing their exact training mix, without which a matched comparison would not have been possible.

References

- [1] Anzalone AV, Randolph PB, Davis JR, et al. Search-and-replace genome editing without double-strand breaks or donor DNA. *Nature* **576**, 149–157 (2019).
- [2] Cao W, Mirjalili V, Raschka S. Rank consistent ordinal regression for neural networks with application to age estimation. *Pattern Recognition Letters* **140**, 325–331 (2020).
- [3] Chen PJ, Hussmann JA, Yan J, et al. Enhanced prime editing systems by manipulating cellular determinants of editing outcomes. *Cell* **184**, 5635–5652 (2021).
- [4] Doench JG, Fusi N, Sullender M, et al. Optimized sgRNA design to maximize activity and minimize off-target effects of CRISPR-Cas9. *Nature Biotechnology* **34**, 184–191 (2016).
- [5] Efron B, Tibshirani RJ. *An Introduction to the Bootstrap*. Chapman & Hall, New York (1993).
- [6] Gu A, Goel K, Ré C. Efficiently modeling long sequences with structured state spaces. *International Conference on Learning Representations* (2022).
- [7] Gu A, Gupta A, Goel K, Ré C. On the parameterization and initialization of diagonal state space models. *Advances in Neural Information Processing Systems* (2022).

- [8] Gu A, Dao T. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv:2312.00752* (2023).
- [9] Hsu JY, et al. OptiPrime: a mechanistic model for prime editing efficiency prediction (2026).
- [10] Kim HK, Kim Y, Lee S, et al. SpCas9 activity prediction by DeepSpCas9, a deep learning-based model with high generalization performance. *Science Advances* **5**, eaax9249 (2019).
- [11] Kim HK, Yu G, Park J, et al. Predicting the efficiency of prime editing guide RNAs in human cells. *Nature Biotechnology* **39**, 198–206 (2021).
- [12] Lieber O, Lenz B, Bata H, et al. Jamba: a hybrid Transformer-Mamba language model. *arXiv:2403.19887* (2024).
- [13] Lorenz R, Bernhart SH, Höner zu Siederdissen C, et al. ViennaRNA Package 2.0. *Algorithms for Molecular Biology* **6**, 26 (2011).
- [14] Mathis N, Allam A, Kissling L, et al. Predicting prime editing efficiency and product purity by deep learning. *Nature Biotechnology* **41**, 1151–1159 (2023).
- [15] Mathis N, Allam A, Tálas A, et al. Machine learning prediction of prime editing efficiency across diverse chromatin contexts. *Nature Biotechnology* (2024).
- [16] Pérez E, Strub F, de Vries H, Dumoulin V, Courville A. FiLM: visual reasoning with a general conditioning layer. *AAAI Conference on Artificial Intelligence* (2018).
- [17] Spearman C. The proof and measurement of association between two things. *American Journal of Psychology* **15**, 72–101 (1904).
- [18] Vaswani A, Shazeer N, Parmar N, et al. Attention is all you need. *Advances in Neural Information Processing Systems* (2017).
- [19] Yu G, Kim HK, Park J, et al. Prediction of efficiencies for diverse prime editing systems in multiple cell types. *Cell* **186**, 2256–2272 (2023).
- [20] Zadrozny B, Elkan C. Transforming classifier scores into accurate multiclass probability estimates. *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (2002).